

Chapter 50: Marine Bioprospecting and Sustainable Utilisation for the Development of Novel Cancer Therapeutics

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Abstract

The world's oceans harbour an extraordinary reservoir of biological diversity that remains comparatively underexploited by modern pharmaceutical science. Marine organisms spanning bacteria, actinomycetes, fungi, microalgae, macroalgae, and mangroves collectively constitute over 90% of oceanic biomass and synthesise a chemically diverse array of secondary metabolites that have evolved primarily as ecological defences against predation, fouling, and microbial competition. Several of these compounds, including polyphenols, polysaccharides, and structurally novel alkaloids and depsipeptides, exhibit pronounced anticancer and antitumour activity through mechanisms such as microtubule stabilisation, inhibition of nucleic acid synthesis, induction of apoptosis, and attenuation of oxidative DNA damage. This chapter provides a systematic account of the principal classes of marine bioresources implicated in cancer therapeutics, critically examines the molecular mechanisms underlying their bioactivity, and situates this discovery pipeline within the broader imperative of sustainable marine resource management, an issue of growing urgency as anthropogenic pressures on coastal and pelagic ecosystems intensify.

Keywords: *diversity, marine bioresources, cancer therapeutics*

1. Introduction to Marine Flora

Marine floras encompassing bacteria, actinobacteria, fungi, and both micro- and macroalgae together constitute approximately 90% of the biomass present in the world's oceans (Sithranga Boopathy & Kathiresan, 2010). This immense and taxonomically diverse biological reservoir is a demonstrated source of medicinally significant chemistry, producing a wide array of bioactive secondary metabolites, most notably polyphenolic compounds and structurally complex polysaccharides.

These marine-derived chemical entities exhibit substantial anticancer and antitumour properties. Beyond their direct cytotoxic action on malignant cells, several have been shown to activate macrophages, induce apoptosis in transformed cells, and attenuate oxidative damage to human DNA, the three converging mechanisms that render them attractive candidates for both cancer chemoprevention and treatment (Sithranga Boopathy & Kathiresan, 2010).

Despite this considerable pharmacological promise, the translational development of marine plant-derived compounds into approved therapeutics remains limited. A great many marine taxa yield bioactive

compounds amenable to isolation and structural characterisation, yet they remain markedly underrepresented within the contemporary pharmaceutical development pipeline. The systematic development of marine floral compounds into validated therapeutic agents is, on the whole, still at a comparatively early stage, leaving considerable scope for further mechanistic research and translational investment.

2. Marine Resources Used in the Treatment of Cancer

Cancer remains among the most formidable challenges confronting modern medicine, and the chemotherapeutic agents currently in clinical use are frequently associated with significant and dose-limiting toxicities. Notably, an estimated 60% of drugs currently approved across therapeutic areas are derived, in whole or in part, from natural sources, underscoring the continued relevance of natural-product-based drug discovery within contemporary pharmacology (Sithranga Boopathy & Kathiresan, 2010).

A broad range of marine resources exhibit anticancer and antiapoptotic properties attributable to the structurally diverse bioactive compounds they produce. Among the secondary metabolites synthesised by marine microorganisms, several have attracted sustained pharmacological interest, including the bryostatins, discodermolide, sarcodictyin, and eleutherobin, which collectively rank among the most extensively studied anticancer agents of marine microbial origin (Sithranga Boopathy & Kathiresan, 2010). Bryostatin 1, first isolated from the bryozoan

Bugula neritina, functions as a potent modulator of protein kinase C and has been shown to induce ubiquitination and proteasomal degradation of the anti-apoptotic protein BCL-2 in lymphoblastic leukaemia cells, while concurrently promoting the proliferation of haematopoietic progenitor cells in the bone marrow.

Marine animal phyla are also recognised producers of toxins, and accumulating evidence suggests that a proportion of the toxins historically attributed to these animals are, in fact, synthesised by associated symbiotic bacteria (Sithranga Boopathy & Kathiresan, 2010). Such microbial toxins have proven valuable in neurophysiological and neuropharmacological research, extending their scientific relevance well beyond oncology alone.

3. Marine Actinomycetes

For more than fifty years, soil-derived actinomycetes of terrestrial origin have constituted a principal pharmaceutical resource for the discovery of antibiotics and structurally related bioactive compounds; marine actinomycetes, by contrast, have received sustained scientific attention only comparatively recently (Sithranga Boopathy & Kathiresan, 2010).

Gutingimycin, a highly polar trioxacarcin derivative, was isolated from a *Streptomyces* species recovered from sediment collected at the Laguna de Términos, Gulf of Mexico (Maskey et al., 2004). The same producer strain has also been reported to yield the structurally related trioxacarcins D–F, in addition to the previously characterised trioxacarcins A–C. More broadly, the family Micromonosporaceae encompasses numerous marine actinomycetes recognised for potent and pharmacologically relevant bioactivity.

A particularly instructive example is thiocoraline, a novel bioactive depsipeptide isolated from *Micromonospora marina*, a marine actinomycete recovered from the Mozambique Strait (Pérez-Baz et al., 1997). Thiocoraline inhibits RNA synthesis and DNA polymerase α activity, arresting the cell cycle in G1 phase, and exhibits selective cytotoxic activity against lung, colon, and renal carcinoma cell lines as well as melanoma (Erba et al., 1999), positioning it as a compelling candidate for further pharmacological development.

4. Marine Bioresource: Marine Bacteria

Marine animals frequently exist in close symbiotic association with marine bacteria, which produce microbial toxins that appear to confer a degree of protection against nervous-system pathologies. This symbiotic relationship has, in turn, become a productive source of pharmacologically active compounds for natural-product researchers (Sithranga Boopathy & Kathiresan, 2010).

One particularly well-characterised example is macrolactin A, a 24-membered macrolide first isolated from an unclassified deep-sea marine bacterium, which inhibits proliferation of B16-F10 murine melanoma cells (Gustafson et al., 1989). Macrolactin A also demonstrates significant inhibitory activity against mammalian herpes simplex virus (HSV) types 1 and 2, and protects T-lymphoblast cells against HIV viral replication, indicating a dual antiviral and antineoplastic relevance (Gustafson *et al.*, 1989).

Cyanobacteria, commonly termed blue-green algae, represent a further well-studied bacterial source. They produce active peptide and polypeptide metabolites that contribute directly to cancer cell death and that modulate intracellular signalling through the activation of protein kinase C (Sithranga Boopathy & Kathiresan, 2010).

5. Marine Bioresource: Marine Fungi

Filamentous marine fungi are frequently of terrestrial origin, with the genera *Penicillium*, *Aspergillus*, and *Fusarium* especially well represented; obligate marine fungi, confined strictly to marine habitats, remain by comparison a largely unexplored resource with considerable untapped biosynthetic potential (Sithranga Boopathy & Kathiresan, 2010).

Biologically active metabolites have been characterised across a range of fungal taxa, including higher basidiomycetes, endophytic fungi, filamentous fungi, and lichens. *Penicillium* species illustrate particularly well how environmental nutrient limitation shapes secondary metabolism: strains deficient in carbon sources tend to favour penicillin biosynthesis, those limited in phosphorus preferentially yield cephalosporin and vancomycin, and those limited in nitrogen tend toward carbapenem production (Sithranga Boopathy & Kathiresan, 2010).

Marine-derived fungi are, more broadly, a well-recognised source of antioxidative natural products, reinforcing their relevance to both cancer prevention and adjunctive therapeutics.

6. Antioxidant Compounds from Marine Fungi

Two antioxidant compounds of particular interest have been isolated from marine fungi: acremonin A, obtained from an *Acremonium* species, and a xanthone derivative recovered from the algicolous fungus *Wardomyces anomalus* (Abdel-Lateff et al., 2003). In the latter case, the xanthone congener 2,3,6,8-tetrahydroxy-1-methylxanthone and the related metabolite 5-(hydroxymethyl)-2-furanocarboxylic acid have both demonstrated significant free-radical-scavenging activity in vitro (Abdel-Lateff et al., 2003).

These compounds are of particular pharmacological interest for their capacity to counteract superoxide and hydroxyl radicals and other reactive oxygen and nitrogen species implicated in a range of chronic diseases, including atherosclerosis, dementia, and cancer. By delaying or preventing oxidative damage, such antioxidants are considered to play a valuable role both in therapeutics and as functional food additives, illustrating the breadth of application that marine-fungal antioxidant chemistry can support.

7. Marine Bioresource: Marine Microalgae

Marine cyanobacteria, or blue-green algae, are considered among the most biosynthetically prolific organisms in the ocean and represent one of the richest known sources of both established and structurally novel bioactive compounds, several of which possess significant pharmaceutical potential (Sithranga Boopathy & Kathiresan, 2010).

Cyanobacteria are also nutritionally notable, producing B-complex vitamins as well as vitamin E. More than half of marine cyanobacterial species are considered potentially exploitable for the extraction of bioactive substances, several of which are effective at inducing apoptosis in cancer cells (Sithranga Boopathy & Kathiresan, 2010).

A well-documented example derives from cell extracts of two photoautotrophic *Calothrix* isolates, which were shown to inhibit, in a dose-dependent manner, the in vitro growth of both a chloroquine-resistant strain of the malaria parasite *Plasmodium falciparum* and human HeLa cervical carcinoma cells (Rickards *et al.*, 1999). Bioassay-directed fractionation of these extracts led to the isolation of calothrixins A and B, pentacyclic metabolites with an indolo[3,2-j]phenanthridine ring system unprecedented among natural products, which exert their growth-inhibitory effects at nanomolar concentrations (Rickards *et al.*, 1999) demonstrating the dual antimalarial and anticancer potential that certain microalgal metabolites possess.

8. Marine Bioresource: Macroalgae

Macroalgae, more commonly known as seaweeds, are marine plants that photosynthesise but reproduce without flowers. They possess a wide range of documented medicinal properties and are an important dietary source of protein, iodine, vitamins, and minerals, and their extracted metabolites have shown promising activity against cancer (Sithranga Boopathy & Kathiresan, 2010).

Edible seaweeds such as *Palmaria palmata* serve as effective antioxidants capable of inhibiting cancer cell proliferation. Similarly, a polysaccharide-rich extract from the brown seaweed *Sargassum thunbergii*, most notably its fucoidan fraction, has demonstrated antitumour activity and inhibition of tumour metastasis in rat mammary adenocarcinoma cells (Itoh *et al.*, 1993), further substantiating the therapeutic promise of macroalgal-derived polysaccharides.

9. Marine Bioresource: Mangroves

Mangroves possess secondary metabolites capable of counteracting cancer and have long been employed within fisherfolk medicine for the treatment of a variety of ailments. On the basis of traditional knowledge combined with preliminary scientific investigation, as many as sixteen mangrove plant species have been identified as candidate sources of anticancer compounds (Sithranga Boopathy & Kathiresan, 2010).

One well-documented example is brugine, a sulphur-containing 1,2-dithiolane alkaloid isolated from the bark of *Bruguiera sexangula*, which displays antitumour activity against sarcoma 180 and Lewis lung carcinoma cell lines (Loder & Russell, 1969). Tannins derived from the same plant have likewise demonstrated anticancer activity against lung carcinoma (Sithranga Boopathy & Kathiresan, 2010), reinforcing the pharmacological value of these coastal ecosystems as a source of structurally diverse anticancer alkaloids.

10. Mechanisms of Anticancer Activity in Marine Resources

Cellular and DNA damage is widely regarded as one of the most important initiating events in carcinogenesis. A reliable biomarker of mutagenic DNA damage is therefore of considerable value both in estimating an individual's cancer risk and in monitoring the efficacy of chemopreventive interventions (Sithranga Boopathy & Kathiresan, 2010).

Under normal physiological conditions, dedicated DNA repair enzymes remove the majority of such lesions. However, oxidative damage to DNA accumulates progressively with advancing age, and cancer risk rises correspondingly. Several coordinated biological mechanisms exist to defend against free radicals and other reactive oxygen species within the human body (Pawlowska et al., 2019).

11. Antioxidant Defence Mechanisms

The body's principal line of defence against oxidative damage is a coordinated enzymatic system comprising superoxide dismutase, glutathione peroxidase, and catalase, complemented by exogenous radical-scavenging micronutrients such as vitamins E and C and various carotenoids (Sithranga Boopathy & Kathiresan, 2010).

Vitamin C is a particularly significant antioxidant, as it is capable of donating a hydrogen atom to form a relatively stable ascorbyl free radical the L-ascorbate anion thereby neutralising more reactive and cytotoxic radical species (Pawlowska et al., 2019). Its role in cancer biology is, however, concentration-dependent: at physiological dietary concentrations vitamin C behaves principally as an antioxidant, whereas at high pharmacological concentrations it can additionally exert pro-oxidant, and potentially tumour-selective, cytotoxic effects (Pawlowska et al., 2019).

12. Immunomodulation and Apoptosis

Apoptosis, or programmed cell death, is a tightly regulated biological process involving numerous intersecting signalling pathways and culminating in a coordinated series of biochemical and morphological changes within the dying cell. The apoptotic machinery is triggered principally through a shift in the balance between pro-apoptotic and anti-apoptotic members of the Bcl-2 protein family (O'Brien & Kirby, 2008).

Dysregulation of apoptotic signalling whether through overexpression of anti-apoptotic proteins such as Bcl-2, or through diminished expression of pro-apoptotic effectors such as Bax and Bak plays a well-established role in disease pathogenesis, and insufficient apoptosis is widely recognised as a contributing factor in the development and progression of cancer (O'Brien & Kirby, 2008). Marine-derived bioactive compounds capable of restoring or enhancing appropriate apoptotic signalling therefore represent a mechanistically grounded and rational strategy for cancer therapeutics.

13. Sustainable Utilisation and Development of Marine Bioresources

The world's oceans function as a principal driver of the global systems that render the Earth habitable for humankind. Drinking water, rainfall, weather, climate, coastlines, and much of the oxygen present in the atmosphere are all provided or regulated, directly or indirectly, by the sea (Secretariat of the Convention on Biological Diversity, 2022).

Careful stewardship of global marine resources is consequently a key feature of any credible sustainable future. At present, however, coastal waters face continuous degradation attributable to pollution and ocean acidification, both of which adversely affect the functioning of marine ecosystems; current estimates indicate that ocean pH has already declined by roughly 30% relative to pre-industrial levels, with cascading and adverse effects on small fish populations that anchor many marine food webs (United Nations Department of Economic and Social Affairs, n.d.).

Protecting the ocean must therefore remain a global policy priority, as marine biodiversity is critical not only to the functioning of marine ecosystems but to human health and planetary wellbeing more broadly (Secretariat of the Convention on Biological Diversity, 2022). Marine protected areas need to be effectively managed and adequately resourced at present, only a small fraction of the ocean is formally protected, well below internationally agreed conservation targets and stronger regulatory frameworks must be implemented to curb overfishing, marine pollution, and ocean acidification if the pharmacological promise of marine bioresources is to be preserved for future discovery and translational use (United Nations Department of Economic and Social Affairs, n.d.).

14. Conclusion

The continuing rise of global warming, malnutrition, and associated environmental pressures is contributing to an increasing global incidence of cancer. Marine plants and microorganisms already possess well-recognised applications within the food and pharmaceutical industries, yet the antioxidant and anticancer activities of a great many marine species remain scientifically unexplored (Sithranga Boopathy & Kathiresan, 2010).

This chapter has surveyed the principal classes of marine bioresources—actinomycetes, bacteria, fungi, microalgae, macroalgae, and mangroves—that have demonstrated anticancer potential, together with the underlying mechanisms of DNA protection, antioxidant defence, and apoptotic regulation through which many of their bioactive compounds are thought to act. Continued research into marine bioprospecting, integrated with a firm commitment to the sustainable stewardship of marine ecosystems, remains essential if this vast and still largely untapped biological reservoir is to be translated into the next generation of effective and less toxic cancer therapeutics.

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